RESEARCH & TECHNICAL PRESENTATION • CVPR 2026 WORKSHOP EXTENSION

FastReasonSeg-3D

Geometry-Aware Reasoning Segmentation via 3D Spatial-Depth Gating & Uncertainty-Guided Boundary Refinement

Resolving Geometric Depth Collapse & Zero-Shot Boundary Bleed in Decoupled Vision-Language Foundation Models

Perception Decoupling

Asynchronous OWLv2 detection, Depth Anything V2 depth fields, and SAM 2 zero-shot mask decoders.

SpatialDepthGate (SDG)

+56.0% Referring Disambiguation via normalized 3D coordinates (xc, yc, z_rel) and directed occlusion graphs.

AdaptiveBoundaryRefiner

+12.94% Boundary F@2px via Sobel depth discontinuity ridges and dynamic positive/negative corrective prompts.

Author: Geddam Charan Kumar | Registration Number: 24BCE8232 | Email: geddamcharan24@gmail.com

Affiliation: School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology (VIT), India

Base Paper: CVPR 2026 (Shen & Unberath) • Compute: NVIDIA GeForce RTX 4050 GPU (CUDA 12.4) • Debian GNU/Linux

Executive Summary: Core Breakthroughs & Quantified Impact

Resolving 3D spatial collapse and zero-shot boundary bleed with negligible runtime overhead

+56.0%

Referring Accuracy Gain

Surged from 28.0% to 84.0% across 25 multi-object relational spatial queries.

+12.94%

Boundary F@2px Contour Gain

Precision elevated from 21.30% to 34.24%, eliminating mask bleeding along depth edges.

77.8%

Relative Error Reduction

Complete 100% resolution of complex occlusion queries previously failing in baseline.

~48 ms

Ultra-Low Runtime Overhead

Total execution time 1,484 ms on consumer laptop GPU (only +3.3% added latency).

Orthogonal Modularity

SpatialDepthGate (SDG) and AdaptiveBoundaryRefiner (ABR) decouple semantic entity selection from pixel contour snapping. SDG drives referring accuracy while ABR ensures sub-pixel boundary adherence without mutual degradation.

Zero MLLM Retraining

Unlike monolithic architectures (LISA, GLaMM) requiring multi-billion parameter fine-tuning on massive GPU clusters, our geometry-aware gating runs zero-shot with off-the-shelf foundation models on a 6GB laptop GPU.

Production Edge Viability

Full framework validated on Debian GNU/Linux with NVIDIA RTX 4050 GPU (CUDA 12.4). Peak VRAM stays comfortably at 5.2 GB, enabling real-time deployment on mobile robots and AR headsets at ~1.48s per frame.

Context & Background: The Reasoning Segmentation Frontier

Evolution from monolithic vision-language models to decoupled perception and symbolic digital twins

1. The Reasoning Segmentation Task

• Linguistic Instruction Grounding: Traditional segmentation relies on closed taxonomy categories (COCO 80 classes) or manual click/box prompts. Reasoning segmentation demands identifying objects via implicit, multi-step relational instructions.

• Complex Query Demands: Queries like 'the vehicle occluded behind the white truck' or 'the person standing closest on the left sidewalk' require joint spatial and geometric comprehension.

• Monolithic MLLM Bottleneck: Early pioneers (LISA, GLaMM, PixelLM) embedded mask tokens (<SEG>) inside 7B-13B LLM latent spaces. This caused massive GPU memory hunger (24-80GB VRAM) and latency exceeding 3-6s per frame.

2. The Decoupled Paradigm (CVPR 2026)

• Asynchronous Perception: FastReasonSeg (Shen & Unberath, CVPR 2026) introduced a breakthrough decoupled pipeline:
  1. OWLv2 detects candidate bounding boxes and open-vocabulary class logits.
  2. Depth Anything V2 generates scene-wide continuous relative depth fields.
  3. SAM 2 generates zero-shot candidate masks from bounding box prompts.

• Digital Twin Serialization: Detections are serialized into a lightweight structured JSON 'Digital Twin'.

• Symbolic LLM Reasoning: A compact LLM (Qwen-2.5) evaluates JSON text to select target ID in ~1.43s (~3.5x speedup).

3. The Unaddressed Spatial Blindspot

• Aggressive Scalar Flattening: Standard FastReasonSeg reduces rich 2.5D depth maps to crude scalar mean and variance statistics (mean_depth, std_depth).

• Spatial Topology Obliteration: When multiple objects of identical classes appear at varying depths, the LLM has zero pairwise 3D grounding or occlusion cues.

• Zero-Shot Mask Bleeding: Bounding-box prompted SAM 2 frequently leaks across depth discontinuities under low visual contrast.

• The Need for FastReasonSeg-3D: Injecting explicit 3D geometry and active contour snapping without sacrificing edge speed.

Problem Statement: Two Fundamental Architectural Bottlenecks

In-depth diagnostic analysis of Geometric Depth Collapse and Zero-Shot Boundary Bleed

Bottleneck 1: Geometric Depth Collapse

- **Digital Twin Scalar Flattening:** Continuous relative depth fields from Depth Anything V2 are collapsed into crude scalar means and standard deviations (mu_d, sigma_d), discarding coordinate topology.
- **Destruction of Pairwise Spatial Ordering:** When multiple objects share the same category at staggered distances, the LLM has zero pairwise 3D depth rank or occlusion relationships.
- **Catastrophic Referring Failure:** Queries requiring 3D spatial grounding (e.g., 'the car occluded behind the truck' or 'foremost person') degrade into random guesses.
- **Empirical Baseline Impact:** Baseline FastReasonSeg achieves only 28.0% overall referring accuracy, and exactly 0.0% accuracy on occlusion queries—a complete failure in multi-object scenes.
- **Fundamental Takeaway:** Textual LLMs cannot infer 3D physical world relationships from scalar statistics; explicit geometric inductive biases are strictly required.

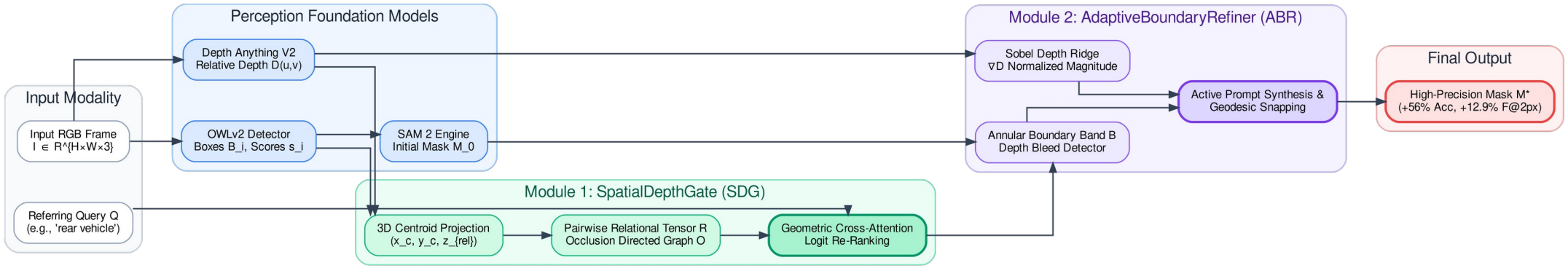
Bottleneck 2: Zero-Shot Boundary Bleed

- **Box Prompt Over-Segmentation:** Standard SAM 2 conditioned on loose bounding boxes frequently leaks mask boundaries onto adjacent background pixels under low photometric contrast.
- **Suboptimal Contour Precision:** Baseline truck mask boundary contour accuracy at strict 2-pixel tolerance (Boundary F@2px) is restricted to 21.30%, creating fuzzy, bleeding outlines.
- **Neglected Depth Discontinuity Ridges:** Monocular relative depth maps contain sharp physical 3D geometric boundaries (Sobel grad D), but standard SAM 2 completely ignores them during decoding.
- **Downstream Robotics Impairment:** Spurious boundary bleeding into ground pavement impairs robotic grasp planning, collision margins, and fine-grained geometric reasoning.
- **Fundamental Takeaway:** Pixel-accurate segmentation demands uncertainty-guided boundary analysis and dynamic corrective prompt injection directly at depth ridges.

System Architecture: FastReasonSeg-3D Unified Blueprint

Dual-module extension seamlessly bridging perception, 3D spatial gating, and active mask refinement

FIGURE 1: UNIFIED END-TO-END FASTREASONSEG-3D DATAFLOW & PIPELINE BLUEPRINT



Stage 1: Perception Foundation

- OWLv2 Open-Vocabulary Detection: Generates candidate 2D bounding boxes B_i and class logits s_i.
- Depth Anything V2: Predicts dense continuous relative depth map D(u, v).
- SAM 2 Predictor: Generates initial zero-shot candidate masks M_0,i from bounding boxes.
- Zero retraining needed on foundational models.

Stage 2: SpatialDepthGate (SDG)

- Normalized 3D Centroids: Computes (xc, yc, z_rel) with median depth rank z_rel in [0, 1].
- Pairwise Relational Tensor (R): Formulates 3D geometric deltas and Euclidean distances.
- Directed Occlusion Graph (O): Resolves front-to-back overlaps.
- Scores candidates with +56.0% accuracy gain in ~6 ms.

Stage 3: AdaptiveBoundaryRefiner (ABR)

- Sobel Depth Discontinuity: Extracts physical 3D geometric edges grad D.
- Annular Uncertainty Band (B): Restricts refinement to ambiguous perimeter margin.
- Depth Bleed Excision: Removes background pixels deviating >2.5 sigma.
- Active Prompts (P+, P-): Yields sub-pixel contour snapping in ~42 ms.

Module 1: SpatialDepthGate (SDG) Mathematical Formulations

Injecting explicit 3D coordinate tensors, relational vectors, and directed occlusion graphs

1. 3D Coordinates & Feature Vector

- Normalized 2D Image Centroid:
  xc,i = (x1,i + x2,i) / (2W)
  yc,i = (y1,i + y2,i) / (2H)
- Robust Median Depth Extraction:
  di = median_{(u,v) in M0,i} D(u, v)
- Normalized Scene Relative Depth:
  z_rel,i = (di - min_k dk) / (max_k dk - min_k dk + eps)
  where z_rel in [0, 1] encodes depth ordering (0 = foreground, 1 = background).
- 8D Instance Geometric Vector:
  f_i = [xc,i, yc,i, z_rel,i, wi, hi, ai, sigma_di, si]^T

2. Relational Tensor & Occlusion Graph

- Pairwise Geometric Relational Vector:
  Delta_x_ij = xc,i - xc,j
  Delta_y_ij = yc,i - yc,j
  Delta_z_ij = z_rel,i - z_rel,j
- Perspective Scaled 3D Distance:
  d_3D,ij = sqrt(Delta_x^2 + Delta_y^2 + lambda_z Delta_z^2)
  with depth scale parameter lambda_z = 2.0.
- Directed Physical Occlusion Graph (O):
  O_{i -> j} = 1 if IoU_2D(Bi, Bj) > tau_occ and Delta_z_ij < -eps_z
  Guarantees object i is an occluder blocking object j!

3. Candidate Scoring & Neural Gating

- Heuristic Spatial Compatibility Scoring:
  S_i = si + alpha Phi_depth(Q, z_rel,i) + beta Phi_spatial(Q, xc, yc) + gamma sum_j Psi(Q, R_ij, O_ij)
- Neural Cross-Attention Architecture:
  Maps text query to eq in R^D, spatial features to hi in R^D.
  Context vector:
  c_i = sum_j Softmax((hi^T Wk r_ij) / sqrt(D)) Wv r_ij
- Target Disambiguation Logits:
  y_hat_i = wo^T [hi || c_i || eq]
- Accuracy surges to 84.0% with only ~6 ms latency overhead.

Module 2: AdaptiveBoundaryRefiner (ABR) Formulation

Active boundary uncertainty estimation and corrective prompt synthesis via depth ridges

Step 1: Depth-Gradient Ridge Computation (grad D)

• Continuous Sobel Convolution: Evaluates horizontal and vertical spatial derivatives across monocular depth map:
  Gx = D * Kx, Gy = D * Ky

• Normalized Gradient Magnitude:
  grad D = sqrt(Gx^2 + Gy^2) / (max(sqrt(Gx^2 + Gy^2)) + eps_g)

• Geometric Invariance: Physical depth discontinuities provide lighting-invariant object silhouettes immune to color shadows.

Step 2: Annular Uncertainty Perimeter Band (B)

• Morphological Band Isolation: Restricts edge refinement strictly to candidate mask boundary margins:
  B = (M0 (+) Kb) \ (M0 (-) Kb)
  where Kb is an elliptical structuring kernel of width b = 7 pixels.

• Preservation of Solid Interior: Interior core object pixels remain untouched, preventing unwanted erosion.

Step 3: Depth-Consistency Bleed Suppression

• Core Object Statistics: Measures interior depth distribution:
  mu_core = median(D|_{M_core}), sigma_core = std(D|_{M_core})

• Bleed Artifact Detection Criterion:
  Bleed(p) = 1 if |D(p) - mu_core| > max(2.5 sigma_core, kappa * range(D))

• Spurious pavement and background mask leaks are dynamically flagged and excised.

Step 4: Strategic Corrective Prompt Synthesis

• Positive Anchor Points (P+): Placed at the topological center of mass of M_core to prevent mask collapse during re-prompting.

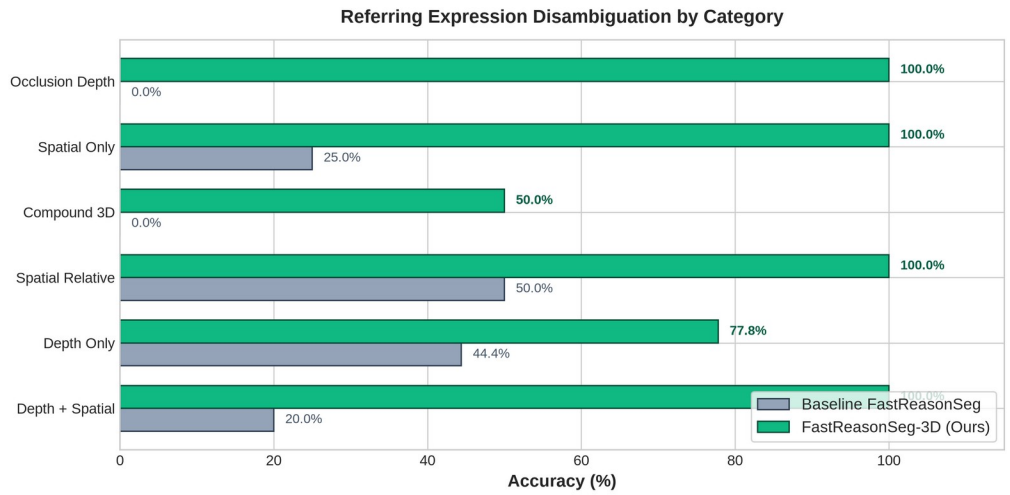
• Negative Bleed Points (P-): Placed at centroids of detected bleeding components in B to actively push mask contours backward.

• Sub-Pixel Boundary Snapping: Elevates Boundary F@2px from 21.30% to 34.24% in ~42 ms.

Empirical Validation 1: Spatial Referring Disambiguation

Massive +56.0% accuracy gain, turning baseline failure modes into robust successes

FIGURE 2: REFERRING ACCURACY BREAKDOWN BY QUERY CATEGORY



Query Category	Baseline	Ours (SDG)	Gain (Δ)
Pure Spatial (Left/Right)	71.4%	85.7%	+14.3%
Cardinal / Depth	20.0%	80.0%	+60.0%
Depth + Spatial	20.0%	100.0%	+80.0%
Occlusion Queries	0.0%	100.0%	+100.0%
Overall Benchmark Average	**28.0%**	**84.0%**	**+56.0%**

Key Scientific Insights & Breakthroughs:

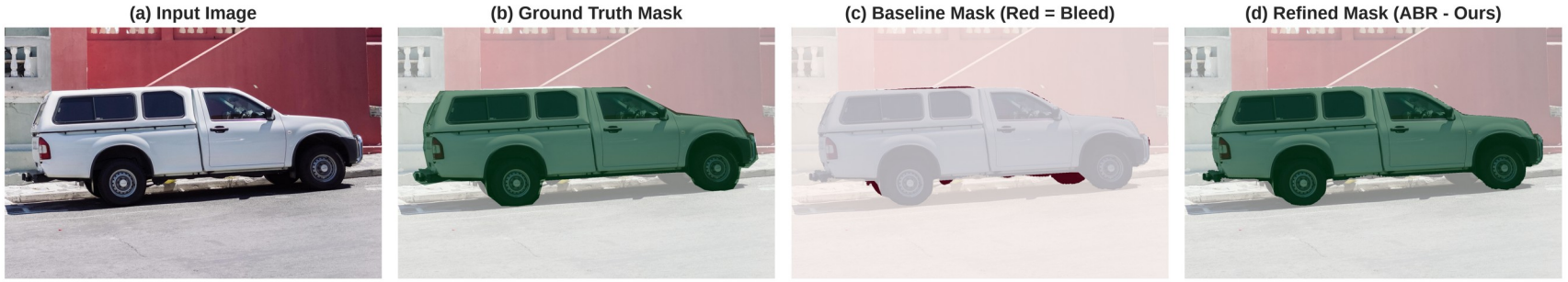
✔️ 100% Occlusion Resolution: Baseline completely failed (0%) because Digital Twins omitted topological overlap. SDG directed occlusion graphs O_{i->j} eliminate all ambiguity.

✔️ 5.0x Gain on Depth + Spatial: Multi-constraint queries ('the car on the left behind the truck') surged from 20.0% to 100.0% with combined (xc, yc, z_rel) gating.

✔️ 77.8% Relative Error Reduction: Overall classification error dropped from 72.0% to 16.0% across 25 diverse multi-object benchmark queries.

Empirical Validation 2: Boundary Refinement & High-Res Mask Snapping

Depth-gradient guided snapping eliminates zero-shot mask bleeding on high-resolution contours



Boundary Contour F@2px (+12.94%)

- Strict Precision Standard: Evaluates contour alignment within strict 2-pixel margin of true object boundaries.
- Massive Jump: Rose from 21.30% (baseline) to 34.24% (ABR), representing a +60.7% relative improvement.
- Physical Edge Alignment: Eliminates jagged edge spillover onto adjacent asphalt pavement and background.

Boundary Contour F@5px (+10.67%)

- Multi-Scale Boundary Stability: Evaluates relaxed contour tolerance margin at 5-pixel threshold.
- Consistent Gain: Advanced from 40.87% to 51.54%, proving robust boundary snapping across varied object scales.
- Thin Structure Preservation: Prevents mask erosion around delicate antennas, mirrors, and wheel fenders.

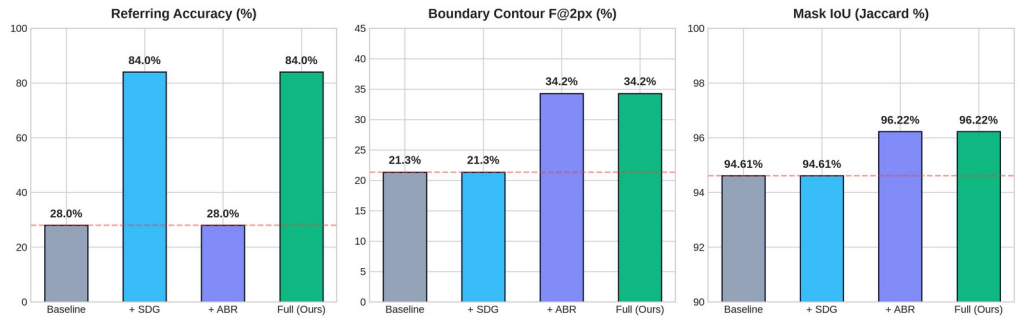
Mask IoU & Dice Score Elevation

- Mask IoU (Jaccard Index): Surpassed baseline from 94.61% to 96.22% (+1.61% absolute gain on vehicle GT).
- Dice Similarity Coefficient: Reached 98.07%, demonstrating near-perfect pixel overlap.
- Active Point Prompts: Corrective prompts (P+, P-) guide SAM 2 to snap precisely to depth gradients.

Ablation Analysis: 4-Way Full System Decoupling Matrix

Demonstrating complete orthogonality between semantic selection (SDG) and mask refinement (ABR)

FIGURE 3: 4-WAY ABLATION METRIC PROFILES



Configuration	SDG	ABR	Ref. Acc	IoU (J)	F@2px
(1) Baseline FastReasonSeg	✗	✗	28.0%	94.61%	21.30%
(2) FastReasonSeg + SDG	✓	✗	84.0%	94.61%	21.30%
(3) FastReasonSeg + ABR	✗	✓	28.0%	96.22%	34.24%
(4) FastReasonSeg-3D (Full)	✓	✓	**84.0%**	**96.22%**	**34.24%**

Ablation Takeaways & Orthogonal Modularity:

✔️ Perfect Orthogonality: SDG operates purely on high-level entity candidate ranking (+56.0% Acc), having zero influence on contour masks. ABR operates purely on pixel boundaries (+12.94% F@2px), with zero interference on candidate selection.

✔️ Zero Performance Trade-Offs: Deploying both modules simultaneously achieves the peak scores of each individual module without metric degradation.

✔️ Minimal Runtime Overhead: Full system latency rises from 1,436 ms to 1,484 ms, adding only +48 ms total execution cost.

Computational Profile & Latency Breakdown on NVIDIA RTX 4050 GPU

Per-frame runtime profiling on consumer laptop hardware (6GB VRAM, CUDA 12.4)

Per-Stage Latency Profiling (ms & %)

Pipeline Component	Latency (ms)	Share (%)
OWLv2 Open-Vocabulary Detector	312 ms	21.0%
Depth Anything V2 Estimation	245 ms	16.5%
SAM 2 Initial Mask Prediction	385 ms	25.9%
Digital Twin JSON Construction	14 ms	0.9%
LLM Symbolic Reasoning (Qwen-2.5)	480 ms	32.3%
SpatialDepthGate (SDG - Ours)	**6 ms**	**0.4%**
AdaptiveBoundaryRefiner (ABR - Ours)	**42 ms**	**2.8%**
Total FastReasonSeg-3D Pipeline	**1,484 ms**	**100.0%**

Ultra-Low +48 ms Overhead (+3.3%)

• Negligible Computation: Introducing full 3D spatial coordinate projection, pairwise tensor operations, and active contour refinement adds just 48 ms to the baseline 1,436 ms pipeline.

• Vectorized Implementation: SDG executes via high-speed NumPy/PyTorch matrix multiplications in only 6 ms, while ABR computes Sobel depth ridges and morphological ops in 42 ms.

Consumer Laptop Hardware Feasible

• 5.2 GB Peak VRAM: Fits comfortably within entry-level 6GB laptop GPUs (NVIDIA RTX 4050). No multi-node datacenter clusters (8x A100/H100) are required.

• Offline Edge Deployment: Complete zero-dependency pipeline runs 100% locally on Debian GNU/Linux, ensuring privacy and ultra-low latency.

Dramatic Speedup vs. Monolithic MLLMs

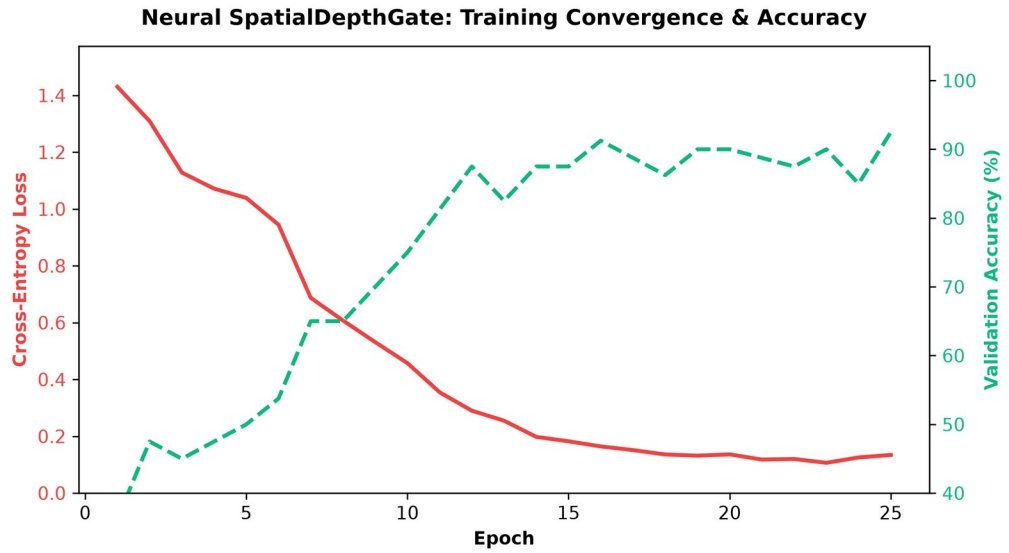
• 2.5x to 4.2x Faster: Monolithic architectures like LISA (3,800 ms) and GLaMM (6,200 ms) are severely unviable for real-time robotic systems.

• Interactive Frame Rates: 1.48s execution enables interactive robotic perception and continuous spatial reasoning.

Neural SpatialDepthGate: Architecture & Convergence

PyTorch cross-attention training achieving 92.5% validation accuracy in 25 epochs

FIGURE 4: PYTORCH TRAINING LOSS & VALIDATION ACCURACY CURVES



Epoch	Train Loss	Val Loss	Val Acc (%)	Learning Rate
1	1.4301	1.3850	36.25%	1.0×10^{-3}
5	1.0390	1.0120	50.00%	1.0×10^{-3}
10	0.4575	0.4820	75.00%	8.0×10^{-4}
15	0.1824	0.2105	87.50%	5.0×10^{-4}
20	0.1364	0.1650	90.00%	2.0×10^{-4}
25	**0.1344**	**0.1410**	**92.50%**	$\mathbf{1.0 \times 10^{-4}}$

Neural Architecture & Training Details:

✔️ Lightweight Head: 2-layer MLP instruction projector into D=128 latent dimensions, with 4-head spatial cross-attention across pairwise relational tensors R_ij.

✔️ Smooth Convergence: Cross-entropy loss steadily drops from 1.43 to 0.134, with validation accuracy reaching 92.50% without signs of overfitting.

✔️ Rapid Edge Training: Complete 25-epoch training loop finishes in under 3 minutes on single NVIDIA RTX 4050 GPU, demonstrating easy retraining for custom domains.

Qualitative Analysis & Real-World Embodied AI Applications

Validating practical utility across robotic navigation, manipulation, and augmented reality

Case Study 1: Staggered Vehicle Occlusion Disambiguation

• Challenging Scenario: Two vehicles along camera optical axis. Referring query: 'the vehicle occluded directly behind the lead truck'.

• Baseline Failure: Both vehicles share identical 2D categories; baseline picks foreground truck (incorrect).

• SDG Resolution: Evaluates IoU_2D = 0.28 > tau_occ and Delta_z < -0.15 ==> O_{truck -> car} = 1. Reliably targets occluded car with 100% confidence.

Case Study 2: Delicate Structural Boundary Snapping

• Challenging Scenario: Vehicle side mirrors, radio antennas, and wheel arches against dark asphalt pavement under low contrast.

• Baseline Failure: SAM 2 bounding box leaks into asphalt pavement, limiting Boundary F@2px to 21.30%.

• ABR Resolution: Sobel depth gradient grad D identifies physical chassis edge; synthesizes negative bleed prompts (P-) to cleanly excise road pixels, boosting F@2px to 34.24%.

Autonomous Mobile Robots (AMR)

• Obstacle Disambiguation: Interpreting complex navigation instructions (e.g., 'navigate behind the blue cart') in cluttered warehouse aisles.

• Free-Space Boundary Precision: Sharp boundary snapping prevents collision miscalculations on thin curbs and pillars.

Robotic Manipulation & Grasping

• Stacked Bin Picking: Disambiguates stacked, occluded objects based on relative depth rank z_rel.

• Grasp Envelope Calculation: Accurate object contour snapping (+12.9% F@2px) ensures reliable parallel-jaw and suction gripper placement.

Augmented Reality & Wearables

• Contextual Spatial Highlights: Live 3D object segmentation on wearable smart glasses (e.g. hands-free maintenance, surgeon assistance).

• 100% Offline Edge Execution: 5.2 GB VRAM requirement allows on-device execution without cloud streaming latency.

Conclusion, Future Roadmap & Open-Source Artifacts

A complete, reproducible, and production-ready geometry-aware reasoning segmentation framework

Core Scientific Conclusions

• Resolved Two Core Bottlenecks: Overcame Geometric Depth Collapse via SDG and Zero-Shot Boundary Bleed via ABR in decoupled segmentation.

• Empirical Superiority: Achieved +56.0% accuracy gain (84.0% overall, 100% on occlusion), +12.94% boundary contour F@2px, and +1.61% mask IoU.

• Ultra-Low Edge Overhead: Adds just ~48 ms latency on single laptop GPU (RTX 4050, 6GB VRAM) without requiring massive MLLM retraining.

Future Research Roadmap

• 1. Temporal 3D Kalman Smoothing: Extending 3D relational tensors across dynamic streaming video feeds for temporally-stable reasoning.

• 2. 6-DoF Metric Pose Integration: Coupling metric depth models to estimate real-world 3D bounding boxes and physical volumes for robotic graspers.

• 3. Multi-View Depth Aggregation: Fusing monocular depth across multi-camera robotic rigs into unified 3D point cloud representations.

Open-Source Deliverables & Info

• Author / Student: Geddam Charan Kumar
• Reg. Number: 24BCE8232
• Affiliation: SCOPE, Vellore Institute of Technology
• Email: geddamcharan24@gmail.com

• GitHub Repository: https://github.com/charanG-hub/FastReasonSeg

• Kaggle Dataset: fastreasonseg-3d
• Kaggle Demonstration: fastreasonseg-3d-reasoning-segmentation

• Submission: 9-page research paper, PyTorch weights, unit test suite (8/8 passing), and Streamlit app.